

Robotic System Implementation Report for AUTONOMOUS SMART CART

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Abstract—This paper presents the design, implementation, and evaluation of an Autonomous Smart Cart system developed using ROS 2 Jazzy and Gazebo Harmonic simulation. The system integrates Ultra-Wideband (UWB) positioning with a 2D LiDAR safety layer to enable a differential-drive robot to autonomously track a user within a simulated indoor retail environment. A UWB simulator node generates four-anchor range measurements from odometry data, a 2D Kalman filter smooths the position estimates, and a proportional follow-me controller drives the cart to maintain a 1.0m following distance. Obstacle safety is enforced by a pass-through node that scales velocity commands based on LiDAR proximity zones before they reach Gazebo. Simulation results in Gazebo Harmonic demonstrate successful follow-me behaviour and layered obstacle avoidance within a simulated indoor retail environment.

I. INTRODUCTION

Shopping malls and large retail environments present significant navigation challenges for users and robotic systems alike. Conventional shopping carts require continuous physical effort and lack intelligent features such as obstacle detection and adaptive movement, making them inefficient and potentially unsafe, particularly for elderly users and parents.

This paper presents the implementation of an Autonomous Smart Cart robotic system, building on the requirements defined in CW1. The focus of CW2 is on realizing key functionalities including follow-me behavior, obstacle avoidance, and sensor integration using ROS2 and Gazebo simulation. The physical hardware prototype has been replaced by a verified Gazebo simulation environment, with RViz used for real-time sensor and navigation visualization.

The system integrates sensing, control, and decision-making to enable safe and efficient navigation in retail environments.

The remainder of this paper is structured as follows: Section II reviews related work; Section III describes the system design and simulation implementation; Section IV outlines non-functional requirements; Section V presents preliminary experimental results; Section VI discusses feasibility; and Section VII presents the testing specifications and risk assessment; Section VIII discusses future improvements; and the appendices cover the use of generative AI and the design thinking process.

II. RELATED WORKS

Shopping malls and large retail environments are designed to accommodate high volumes of customers, often resulting in crowded aisles, dynamic pedestrian flow, and limited maneuvering space. In such environments, conventional shopping carts present operational challenges including limited

maneuverability, reduced braking control under heavy loads, and increased physical strain on users. These challenges are particularly significant for elderly shoppers and parents managing children, where divided attention and physical effort increase the risk of minor collisions and unsafe interactions [1], [2].

Studies in retail ergonomics show that pushing heavily loaded carts significantly increases user exertion and reduces mobility efficiency, particularly among elderly individuals [3]. Additionally, pedestrian flow research shows that dense crowd conditions require constant micro-adjustments in direction and speed to avoid collisions, increasing cognitive load and reaction time demands [4], [5]. These conditions can lead to sudden stops, braking instability, and unintended contact with other shoppers.

From a robotics perspective, indoor environments such as shopping malls present complex navigation challenges. Unlike structured industrial settings, retail environments are highly dynamic, with unpredictable human motion, temporary obstacles, and occlusions [6]. Human-robot interaction (HRI) research emphasizes the importance of adaptive speed control, obstacle prediction, and intuitive user feedback mechanisms to ensure safety and social acceptance in shared environments [1].

Furthermore, collision-related incidents involving shopping carts, particularly heel and ankle impacts, are commonly associated with poor braking response and momentum under heavy load [7], [8]. This highlights the need for intelligent control systems capable of adjusting speed dynamically based on load conditions and crowd density.

To address these challenges, this project proposes a Smart Autonomous Shopping Cart Robot designed to aid users through adaptive navigation, obstacle detection, human-aware motion control, and improved ergonomics. The system aims to enhance safety, reduce physical strain, and improve shopping efficiency in crowded retail environments.

Building on this foundation, this paper presents the simulation-based implementation of the Smart Cart using ROS2 and Gazebo, validating the proposed control architecture in a physics-accurate virtual retail environment.

III. SYSTEM DESIGN AND SIMULATION IMPLEMENTATION

A. System Overview

The Smart Cart Robot is an autonomous indoor assistive system designed to improve shopper safety and efficiency. It operates independently but relies on mall layout data for

navigation and real-time sensing to detect obstacles, crowded areas, and user location. The robot interacts with users via visual and auditory feedback and is designed to complement traditional shopping carts, not replace them entirely. It is a self-contained system with integrated mobility, sensing, and control subsystems.

The Smart Cart performs autonomous navigation, follow-me tracking, obstacle avoidance, and real-time user feedback within indoor retail environments.

The Smart Cart system is designed to serve a broad range of shoppers in retail environments. Three distinct user classes have been identified, each with different interaction needs and system expectations.

The system targets three user groups: general shoppers, elderly users, and parents with children. Each group requires different levels of automation, safety response, and interaction simplicity

The Smart Cart Robot is intended for use in indoor retail environments such as malls or supermarkets. It is designed to operate on flat floors with occasionally mild ramps, in moderate lighting conditions, and among shoppers, carts, and shelves. It is optimized for environments with varying crowd densities and narrow aisles, where human safety and maneuverability are critical.

B. Robot Model

Design constraints for the Smart Cart Robot include:

- Maximum cart width: 0.7 m to navigate standard aisles
- Maximum weight: 40 kg including battery and electronics
- Safety mechanisms to prevent collisions and injuries
- Battery must provide 2–3 hours of continuous operation
- Cost and materials suitable for scalable deployment
- Limited integration with external IT or payment systems

The Smart Cart hardware integrates a layered architecture of sensing, actuation, power, and control subsystems. All components are mounted on or within the existing cart frame to minimize structural modifications. The Raspberry Pi 4B serves as the high-level processing unit running ROS 2, while the Arduino Mega handles real-time low-level motor control and sensor I/O.

C. System Architecture Overview

This diagram illustrates the **Autonomous Smart Cart System** architecture, divided into high-level intelligence and low-level execution systems.

At the top tier, a **Raspberry Pi 4** serves as the system’s “brain,” running a **ROS 2 stack** to manage complex computational tasks like SLAM (Simultaneous Localization and Mapping), UWB-based positioning, and path planning for realizing movement. This high-level controller communicates via USB Serial with an **Arduino Mega**, which functions as the low-level “nervous system.” The Arduino is dedicated to real-time hardware interfacing, utilizing PWM signals to command a quad H-Bridge motor driver system and managing a diverse sensor suite including infrared, ultrasonic, and weight sensors through Serial UART protocol.

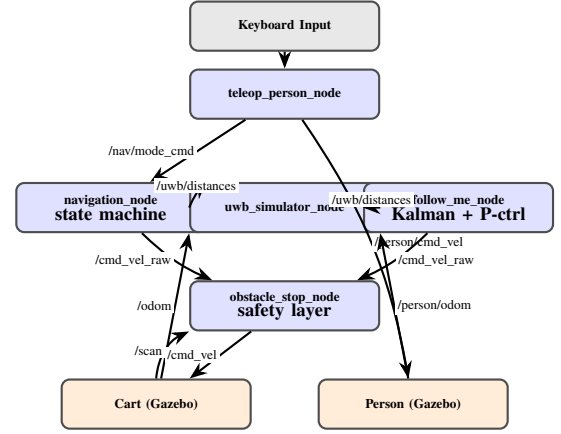


Figure 1. ROS 2 Software Architecture — Node and Topic Data Flow (Author Developed)

Supporting this control structure is a robust physical and power foundation designed for mobility, safety and endurance. The drive system employs **four high-torque DC motors** paired with **omni-wheels** and encoders, enabling precise, multi-directional movement. The entire assembly is energized by a high-capacity **24 V 20 Ah LiPo battery**, which is regulated down to 5 V and/or 3.3 V to safely power the microelectronics. A dedicated Battery Management System (BMS) ensures electrical safety and longevity, creating a reliable hardware closed-loop that translates high-level navigation commands into smooth, physical locomotion.

D. Drive and Steering Mechanism

The cart employs four omnidirectional (omni) wheels driven by independent high-torque DC motors, enabling holonomic motion — movement in any direction without requiring the cart to rotate first. This is critical for navigating narrow supermarket aisles and performing lateral parking maneuvers.

Table I
DRIVE SYSTEM SPECIFICATIONS (AUTHOR DEVELOPED)

Component	Specification	Justification
Omni Wheels (x4)	Diameter: 150 mm, Load: 15 kg each	Holonomic motion; lateral movement in aisles
DC Motors (x4)	12 V, 150 RPM, 20 kg-cm torque	Sufficient torque for 40 kg loaded cart
Motor Driver	L298N Dual H-Bridge ×2	PWM speed + direction control per motor
Encoders	Quadrature, 500 PPR	Closed-loop odometry for position tracking
Max Speed	1.2 m/s (unconstrained)	Capped at 0.8 m/s in software near humans
Turning Radius	0 cm (in-place rotation)	Full holonomic capability via omni-wheels

Note: The physical design uses omni-wheels for holonomic motion. In the Gazebo simulation a standard differential-drive model is used, as the follow-me and obstacle avoidance algorithms do not require holonomic kinematics to be validated. The full omni-wheel model would be integrated in the physical prototype.

1) *Mechanical Design Calculations:* The mechanical design of the Smart Cart must ensure that the selected wheel and motor configuration is capable of moving the cart safely and reliably under expected operating conditions. Mechanical feasibility is verified using rolling resistance and torque analysis.

For the proposed design, the maximum operational cart mass is limited to **40 kg**, including the cart structure, battery, electronics, and payload. Since the cart is intended for indoor use on smooth retail flooring, the dominant resistive force during motion is rolling resistance. For hard wheels operating on tiled or polished mall floors, a rolling resistance coefficient of approximately **0.02** can be assumed.

The rolling resistance force is estimated using:

$$F_r = \mu_r \times m \times g \quad (1)$$

where F_r is the rolling resistance force (N), μ_r is the rolling resistance coefficient, m is the total cart mass (kg), and g is gravitational acceleration = 9.81 m/s^2 .

Substituting the design values:

$$F_r = 0.02 \times 40 \times 9.81 = 7.848 \text{ N} \quad (2)$$

Thus, the approximate force required to maintain motion on a flat indoor floor is **7.85 N** under ideal rolling conditions.

To determine the torque required at the wheel:

$$T = F \times r \quad (3)$$

where T is torque (Nm), F is the force at the wheel-ground interface (N), and r is the wheel radius (m).

The selected omni wheel diameter is **150 mm**, giving a radius of:

$$r = 75 \text{ mm} = 0.075 \text{ m} \quad (4)$$

Therefore, the required wheel torque is:

$$T = 7.848 \times 0.075 = 0.589 \text{ Nm} \quad (5)$$

The selected motor specification is **20 kg-cm torque**. Converting to SI units:

$$20 \text{ kg-cm} = 20 \times 0.0981 = 1.962 \text{ Nm} \quad (6)$$

Hence, each selected motor is capable of delivering approximately **1.96 Nm** torque. Since the required torque per wheel for steady-state motion is approximately **0.59 Nm**, the motor provides more than three times the required torque under nominal conditions. This gives a sufficient design margin for additional real-world effects such as wheel slip, uneven flooring, transient acceleration, minor ramps, misalignment losses, and increased resistance when the cart starts moving from rest.

The linear speed of the cart can also be checked using the selected motor speed of **150 rpm**. The wheel circumference is:

$$C = \pi D = \pi \times 0.15 = 0.471 \text{ m} \quad (7)$$

The theoretical linear speed is therefore:

$$v = \frac{150 \times 0.471}{60} = 1.18 \text{ m/s} \quad (8)$$

This closely matches the system specification of approximately **1.2 m/s maximum unconstrained speed**. However, in practical operation, software speed limiting reduces the cart

speed to approximately **0.8 m/s** near users to improve safety in human-populated indoor spaces.

Overall, the calculations indicate that the selected wheel and motor combination is mechanically adequate for the intended payload and indoor operating environment.

Omni wheels were selected to enable holonomic motion, allowing lateral movement and zero-radius turning. This improves navigation in narrow aisles and crowded environments compared to differential drive systems.

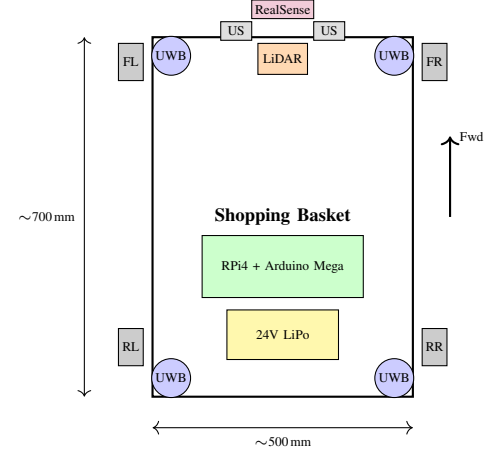


Figure 2. Mechanical System Diagram — Top-Down Layout (Author developed)

2) *Mechanical System Diagram:* The mechanical system of the Smart Cart is based on a compact rectangular chassis supporting the shopping basket, battery system, sensor mounts, and control electronics. The load-bearing structure is designed to distribute mass evenly across four independently driven omni wheels located at the corners of the base frame. This arrangement provides both mechanical stability and holonomic mobility.

A practical top-level mechanical layout is described as follows:

- The **shopping basket** is mounted centrally above the chassis frame.
- Four **150 mm omni wheels** are mounted at the four corners of the base.
- Each wheel is coupled to an independent **12 V DC geared motor**.
- The **battery pack** is placed near the lower centre of the chassis to reduce the centre of gravity.
- The **motor drivers, Arduino Mega, and Raspberry Pi** are mounted in a protected electronics enclosure below or behind the basket.
- **Ultrasonic sensors** are positioned on the front face of the cart for obstacle detection.
- **IR proximity sensors** are mounted near the lower perimeter for foot-level and near-ground detection.
- **Load cells** are integrated beneath the basket support structure, one near each corner, to monitor payload distribution and total load.
- A **remote receiver / UWB anchors** are mounted around the cart frame to maintain robust user localisation perfor-

mance.

From a structural perspective, the chassis acts as the main load-transferring member between the basket and the wheels. The wheelbase dimensions must remain compact enough to navigate aisles while still providing adequate resistance to tipping. The frame must also withstand static and dynamic loads generated during payload carrying, acceleration, deceleration, and emergency stopping.

3) *Stability Analysis*: Stability is ensured through low center of gravity, wide wheel placement, and speed limiting. The system reduces speed under high load and halts operation if tilt exceeds safety thresholds.

4) *Braking and Mechanical Safety*: The braking system uses controlled deceleration, motor braking, and hardware emergency cut-off. Speed limits and load-aware control ensure safe stopping distances in indoor environments.

E. Sensor System

1) *2D Lidar*: A 2D Lidar (e.g., RPLIDAR A1) is mounted on the front-top of the chassis to provide a 360° horizontal scan of the environment. This allows the cart to detect obstacles beyond the narrow cone of ultrasonic sensors and facilitates SLAM (Simultaneous Localization and Mapping).

2) *Depth Camera (Human Pose Estimation)*: An RGB-D camera (e.g., Intel RealSense) is mounted on the handle assembly to provide 3D environmental data. Unlike the UWB which tracks the user, the camera is dedicated to Visual Odometry. By tracking “features” in the mall environment, it provides a secondary position estimate that prevents “drifting” in areas where UWB signals may experience multipath interference from metallic shelving.

3) *UWB Positioning (Indoor Localization)*: Four Ultra-Wideband (UWB) receiver anchor modules are mounted at the corners of the cart frame (RL, FL, RR, FR as shown in the design diagram). The user carries a UWB transmitter tag embedded in the handheld remote. Time-Difference-of-Arrival (TDoA) calculations between the anchors provide sub-30 cm indoor positioning accuracy, enabling reliable follow-me tracking even when the user is briefly occluded. Power Management System.

Note: In the Gazebo simulation, UWB positioning is replaced by direct odometry from the avatar model. The avatar’s /odom topic serves as the ground-truth user position input to the follow-me controller, replicating the role of UWB in the physical design.

4) *Ultrasonic Obstacle Detection*: Two HC-SR04 ultrasonic sensors are mounted on the front face of the cart basket at mid-height to detect obstacles — including other shoppers, shelves, and stray items — within a forward-facing cone. Readings are processed by Arduino and published to ROS 2 via serial bridge.

5) *IR Sensors (Proximity / Edge Detection)*: Infrared sensors are placed at the lower perimeter of the base platform (front and sides) to detect low-lying obstacles such as children’s feet, bags on the floor, and kerb edges. IR sensors also function as a secondary proximity check at close range where ultrasonic sensors may have dead zones.

6) *Weight Sensors (Load Monitoring)*: Four load cell weight sensors are embedded beneath the basket floor, one at each corner. Total basket load is used to dynamically adjust motor torque limits and maximum speed — heavier loads reduce maximum speed to maintain braking safety. A HX711 ADC module interfaces the load cells to the Arduino.

Table II
SENSORS USED FOR ROBOT MODEL

Sensor	Purpose
LiDAR	Mapping & obstacle detection
Depth Camera	Visual odometry
UWB	User tracking
Ultrasonic	Front obstacle detection
IR	Low-level obstacle detection
Load Cells	Weight-based speed control

7) *RGB LED Status Indicators*: A strip of RGB LEDs is mounted along the top rail of the cart basket, visible at standing height. LEDs provide real-time status feedback to both the user and nearby shoppers without requiring audio in noise-sensitive environments.

Table III
RGB LED STATUS COLOR CODING (AUTHOR DEVELOPED)

Color	State	Meaning
Green (slow pulse)	Following user	Normal follow-me operation
Blue (solid)	Idle / Parked	Waiting for user command
Yellow (flashing)	Obstacle detected	Slowing down, path adjusting
Red (rapid flash)	Emergency stop	Obstacle <30 cm, brakes applied
White (solid)	Charging	Connected to charging station

F. Handheld Remote Control

A compact handheld remote is provided to the user at the start of each shopping session. It contains the UWB transmitter tag for positioning and two physical buttons. The remote communicates with the cart’s receiver module over the UWB link.

Table IV
HANDHELD REMOTE SPECIFICATIONS (AUTHOR DEVELOPED)

Control	Function	Behavior
F (Follow) Button	Activate follow-me mode	Cart follows user at 0.8–1.0 m distance
X (Stop) Button	Emergency / manual stop	Immediately cuts motor power, brakes applied
UWB Tag (embedded)	Real-time user location	Transmits position to 4 onboard anchors

G. Power Management System

The cart is powered by a central 24 V lithium polymer (LiPo) battery pack with a dedicated Battery Management System (BMS) for protection against over-discharge, overcharge, and

thermal runaway. Step-down regulators supply stable 5 V and 3.3 V rails to the Raspberry Pi, Arduino, and sensors respectively.

Table V
POWER MANAGEMENT SYSTEM SPECIFICATIONS (AUTHOR DEVELOPED)

Component	Specification	Purpose
Main Battery	24 V, 20 Ah LiPo (480 Wh)	Primary motor and electronics power
BMS	24 V 30 A balancing BMS	Cell balancing, over-current protection
Buck Regulator 1	24 V $\rightarrow$ 5 V, 10 A (XL4016)	Raspberry Pi 4B, USB devices
Buck Regulator 2	5 V $\rightarrow$ 3.3 V, 3 A (AMS1117)	Sensors and Arduino 3.3 V peripherals
Motor Power Rail	24 V direct (via BMS)	4 $\times$ DC motors via H-Bridge drivers
Battery Life	$\sim$ 3 hours at avg load	Full shopping session requirement met
Charge Connector	XT90 + CC/CV charger	Return-to-base charging protocol

H. Safety and Fault Detection Mechanisms

Hardware-level safety mechanisms operate independently of the software stack to guarantee failsafe behavior even in the event of a software crash or sensor fault.

Table VI
HARDWARE SAFETY AND FAULT DETECTION MECHANISMS (AUTHOR DEVELOPED)

Safety Feature	Trigger	Condition	Response
Hardware E-Stop	X button on remote		Direct relay cut to motor drivers
Obstacle Brake	Ultrasonic	<30 cm	PWM zeroed, regenerative braking
Watchdog Timer	Arduino no heartbeat (>500 ms)		Motors disabled, LED red flash
Battery Low	<20% charge (BMS signal)		Speed limited to 50%, audible beep
Overload	Basket >40 kg		Cart stops, user notified via LED
Tilt/Fall Detection	IMU	pitch/roll >15°	Motors disabled, alert triggered

I. Software Overview

Note: The hardware design described in Sections III-A through III-F covers the full physical system specification. The Gazebo simulation validates the core navigation algorithms (UWB follow-me, Kalman filtering, obstacle avoidance). Hardware-specific components — load cells, RGB LEDs, depth camera, and Arduino serial bridge — are part of the physical design and are not instantiated in simulation.

The Smart Cart software stack is built on ROS 2 (Jazzy Jalopy) and simulated within Gazebo. The architecture follows a modular node-based design, where each functional subsystem is implemented as an independent ROS 2 node communicating via a publish-subscribe model.

In the simulation, a unified teleop node serves dual purposes: WASD keyboard input drives the user avatar model in Gazebo,

while keys 1, 2, and 3 simulate the handheld remote, publishing mode commands to three topics: `/remote/button` (String label), `/remote/button_int` (integer code), and `/nav/mode_cmd` (String command). Keys 1, 2, 3 map to STOP, FOLLOW, and IDLE respectively. This allows a single operator to control both the person avatar and the cart mode during simulation testing.

J. Layered ROS 2 Architecture

- `gz_sim` — Gazebo Harmonic physics engine; hosts cart and person models, publishes sensor data via ROS-Gz bridge
- `robot_state_publisher` — publishes URDF transforms for the cart on `/tf`
- `joint_state_publisher` — publishes cart joint states
- `teleop_person_node` — unified keyboard input node; WASD drives the person avatar on `/person/cmd_vel`; keys 1, 2, 3 publish Stop/Follow/Idle commands to `/nav/mode_cmd`
- `uwb_simulator_node` — simulates four UWB anchors using cart and person odometry; publishes distances on `/uwb/distances` (Float32MultiArray) and RViz markers on `/uwb/markers`
- `navigation_node` — state machine managing IDLE, FOLLOW, STOP, and MANUAL modes; subscribes to `/nav/mode_cmd` and publishes `/nav/current_mode` and `/nav/status`; includes a 5-second UWB watchdog
- `follow_me_node` — UWB trilateration + Kalman2D filter; proportional controller publishes velocity commands to `/cmd_vel_raw`
- `obstacle_stop_node` — subscribes to `/cmd_vel_raw` and `/scan`; scales velocity based on LiDAR proximity and publishes to `/cmd_vel`
- `rviz2` — real-time visualisation of LiDAR scan, cart pose, UWB markers, and person position

K. Gazebo Simulation Environment

The Smart Cart system is implemented and tested within a Gazebo simulation environment replicating an indoor retail space. The simulated world includes aisle structures, static shelf obstacles, and a mobile user agent for follow-me validation.

1) *Robot Model (URDF)*: The cart is modeled as a differential-drive robot in URDF format, with the following key elements:

- Base chassis link with inertial mass of 15 kg (robot body); maximum payload capacity is 40 kg
- Four wheel joints (front-left, front-right, rear-left, rear-right) with continuous rotation
- GPU LiDAR sensor (`gpu_lidar` type) mounted at chassis front-top; bridged to `/scan` via `ros_gz_bridge`
- Differential drive plugin (`gz-sim-diff-drive-system`, Gazebo Harmonic) subscribing to `/cmd_vel`
- Four UWB anchor visual links (`uwb_fl`, `uwb_fr`, `uwb_rl`, `uwb_rr`) at cart corners

- Intel RealSense-style depth camera (`camera_link`) bridged to `/camera/color/image_raw` and `/camera/depth/image_raw`

2) *Gazebo World Configuration*: The simulated retail environment consists of:

- Supermarket world with left/right walls (± 2.5 m), two shelf models (± 1.5 m), and a static orange obstacle box ($0.3 \times 0.3 \times 0.5$ m) at position (2.0, 0.6) to simulate aisle obstructions
- The cart spawns at origin (0, 0, 0.20 m); the person model spawns at (2.0, 0.0, 0.12 m)
- The person is a diff-drive robot model with human visual geometry (head, torso, arms, legs) receiving velocity commands on `/person/cmd_vel` and publishing odometry on `/person/odom`
- Spawn positions and launch timing defined in `start_all.launch.py` using staged `TimerAction` delays (Gazebo $\rightarrow t=0$, cart spawn $\rightarrow t=3$ s, person $\rightarrow t=4$ s, ROS nodes $\rightarrow t=6$ s, RViz $\rightarrow t=7$ s)

3) *ROS2 Node Graph*: The system is implemented in ROS 2 using modular nodes for perception, localization, and control. Communication between Raspberry Pi and Arduino is handled via a serial bridge for real-time actuation.

The active ROS2 nodes during simulation include:

- `/gazebo` — physics simulation, sensor publishing, and avatar/cart model hosting
- `/robot_state_publisher` — URDF transforms for cart and user avatar
- `/teleop_node` — unified keyboard input node; WASD controls user avatar movement; keys 1, 2, 3 send remote commands (Stop, Follow, Idle) to the cart
- `/follow_me_controller` — subscribes to avatar pose and publishes `/cmd_vel` to cart when in Follow mode
- `/obstacle_detector` — LiDAR-based proximity zone classifier
- `/rviz2` — real-time visualization of LiDAR scan, cart pose, and avatar position

The two-stage velocity pipeline is a key architectural feature: `follow_me_node` computes velocity and publishes to `/cmd_vel_raw`; `obstacle_stop_node` intercepts this, applies proximity-based scaling, and forwards the result to `/cmd_vel` which Gazebo’s `DiffDrive` plugin consumes. This ensures obstacle safety is enforced regardless of the follow-me controller’s output.

L. Localization Subsystem

User localisation in simulation is achieved through a UWB simulation and Kalman filtering pipeline implemented across two nodes.

The `uwb_simulator_node` places four virtual anchors at the cart corners. At each timestep (20 Hz), it reads the current Gazebo positions of both the cart (`/odom`) and the person (`/person/odom`), computes the Euclidean distance from each anchor to the person, and adds Gaussian noise ($\sigma = 0.08$ m, bias = 0.02 m) to simulate realistic UWB

hardware error. These four distances are published as a `Float32MultiArray` on `/uwb/distances`.

The `follow_me_node` consumes these distances and runs a least-squares trilateration algorithm to estimate the person’s 2D position relative to the cart. This raw estimate is fed into a `Kalman2D` filter (from `uwb_filter_node.py`) with process noise $Q = 0.05\mathbf{I}$ and measurement noise $R = 0.3\mathbf{I}$. The filter has a 10-cycle warmup period during which the raw trilateration result is used directly. After warmup, the smoothed estimate drives the follow-me proportional controller.

Table VII
LOCALISATION SUBSYSTEM ROS 2 TOPICS (AUTHOR DEVELOPED)

Topic	Message Type	Description
<code>/uwb/distances</code>	<code>Float32MultiArray</code>	Noisy distances from 4 UWB anchors to person
<code>/uwb/distance</code>	<code>Float32</code>	Scalar distance to person
<code>/uwb/angle</code>	<code>Float32</code>	Bearing angle to person (rad)
<code>/uwb/markers</code>	<code>MarkerArray</code>	RViz visualisation of UWB geometry
<code>/odom</code>	<code>Odometry</code>	Cart wheel odometry
<code>/person/odom</code>	<code>Odometry</code>	Person model odometry (ground truth)
<code>/scan</code>	<code>LaserScan</code>	2D LiDAR data for obstacle detection

M. Obstacle Detection and Human Tracking

The perception node implements a multi-modal approach by fusing data from the 2D LiDAR, Depth Camera, Ultrasonic, and IR sensors. The 2D LiDAR provides a high-resolution `/scan` topic used for long-range (up to 12 m) obstacle mapping and detection of static objects like shelves. The Depth Camera identifies dynamic obstacles and filters out “noise” (such as overhead signage) by focusing on the 3D space directly in the cart’s path.

The `obstacle_stop_node` monitors only the forward $\pm 45^\circ$ arc of the LiDAR scan. Obstacles are classified into three proximity zones and a speed scale factor applied to the incoming `/cmd_vel_raw` velocity before forwarding to `/cmd_vel`: CAUTION (< 1.5 m, 80% speed), SLOW (< 0.8 m, 50% speed), and EMERGENCY (< 0.3 m, 0% speed). In FOLLOW mode, LiDAR returns closer than 0.4 m are ignored to prevent false stops triggered by the followed person.

Table VIII
OBSTACLE PROXIMITY ZONES AND CART RESPONSES (AUTHOR DEVELOPED)

Zone	Distance	Behaviour
NORMAL	≥ 1.5 m	Full speed (100%)
CAUTION	< 1.5 m	Speed reduced to 80%
SLOW	< 0.8 m	Speed reduced to 50%
EMERGENCY	< 0.3 m	Full stop (0%), hold position

N. Follow-Me Controller

The follow-me controller in `follow_me_node` implements a pure **proportional (P) controller** — there is no path

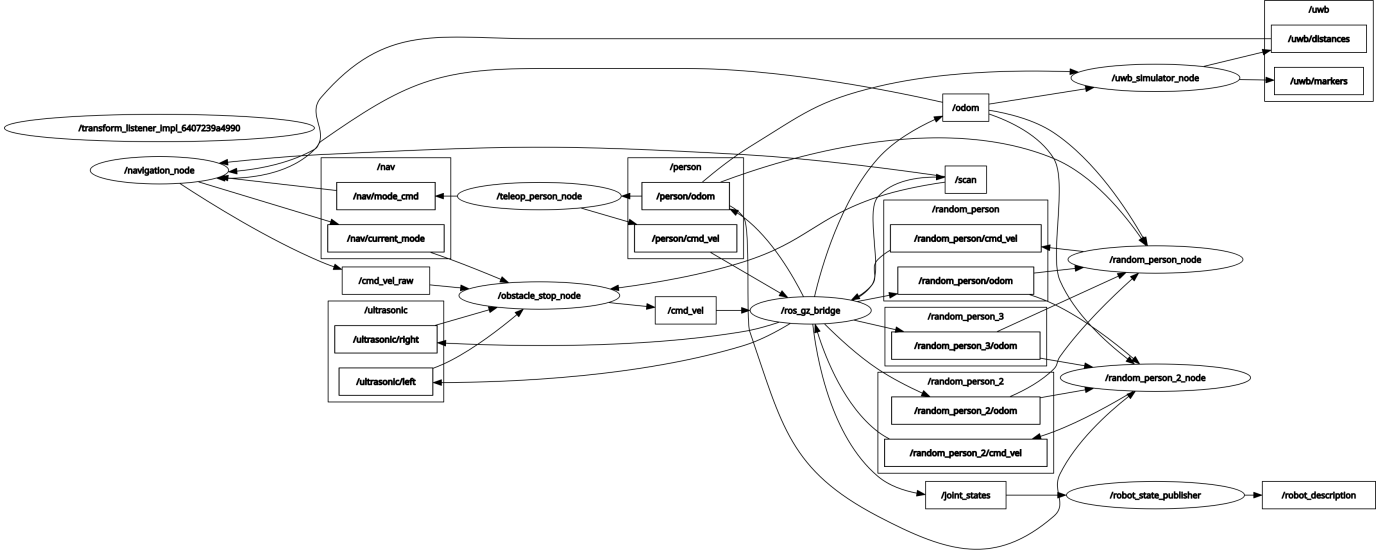


Figure 3. ROS 2 Node Graph from rqt_graph — Gazebo Simulation (Author developed)

planner, costmap, or DWA. The controller runs at 20 Hz and computes:

$$v_{linear} = \min(v_{max}, K_P \cdot e_{dist}) \quad (9)$$

$$\omega_{angular} = K_P \cdot e_{angle} \quad (10)$$

where $K_P = 0.6$, $v_{max} = 0.8$ m/s, e_{dist} is the error from the 1.0 m target following distance, and e_{angle} is the bearing angle to the person. A dead zone of 0.15 m suppresses linear motion when the cart is within tolerance, and an angle deadband of 0.15 rad suppresses small angular corrections.

If no UWB data arrives within 1 second, the controller stops the cart. The `navigation_node` watchdog additionally reverts the mode to IDLE if the UWB link is lost for more than 5 seconds.

O. Arduino–Raspberry Pi Communication Bridge

The Arduino Mega runs dedicated firmware that handles all real-time sensor sampling and motor PWM generation. It communicates with the Raspberry Pi over USB serial at 115200 baud using a lightweight custom binary protocol. The `sensor_bridge` ROS 2 node on the Raspberry Pi handles serial parsing and publishes all sensor data to appropriate topics. A 500 ms watchdog timer on the Arduino will cut motor power if no heartbeat is received from the Raspberry Pi, preventing runaway behavior during software crashes.

P. End-to-End Operational Flowchart

The following flowchart shows the complete operational logic of the Smart Cart Robot from power-on through to session end. It covers normal follow-me operation, obstacle avoidance decision-making, load-based speed adaptation, and emergency stopping.

Table IX
ARDUINO–RASPBERRY PI SERIAL COMMUNICATION PROTOCOL (AUTHOR DEVELOPED)

Message Type	Direction	Frequency	Content
SENSOR_DATA	Arduino → RPi	50 Hz	Ultrasonic, IR, weight, encoder vals
MOTOR_CMD	RPi → Arduino	50 Hz	PWM duty cycle for 4 motors
LED_CMD	RPi → Arduino	5 Hz	RGB colour and blink pattern
HEARTBEAT	RPi → Arduino	10 Hz	Watchdog keepalive signal
STATUS_ACK	Arduino → RPi	10 Hz	Fault flags, battery level

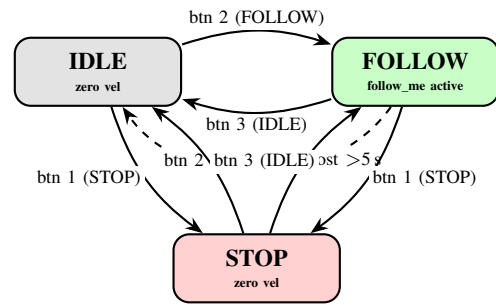


Figure 4. Navigation State Machine — Mode Transitions (Author Developed)

Q. Assumptions and Dependencies

Assumptions and dependencies for the Smart Cart Robot include:

- Mall layout maps are available and accurate for navigation.
- Floors are smooth with minor slopes only.
- Users are able and willing to interact with the cart as instructed.
- Simulated sensors and actuators in Gazebo approximate real hardware behavior within acceptable tolerances.

- Gazebo physics parameters are tuned to reflect indoor retail floor conditions.

IV. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality standards and operational constraints the Smart Cart system must meet — not what it does, but how well it does it. These requirements directly influence the user experience, safety, and deployment viability of the system.

A. Performance Requirements

The system must meet the following performance thresholds to function safely and usefully in a real retail environment:

Table X
NON-FUNCTIONAL REQUIREMENTS (AUTHOR DEVELOPED)

Requirement	Target Value	Justification
Sensor-to-decision latency	<30 ms	Cart must react to obstacles before reaching unsafe proximity at walking speed
Emergency stop response time	<200 ms	Heel-impact prevention requires sub-200 ms full brake from walking speed
UWB position update rate	20 Hz	Simulated UWB node publishes at 20 Hz; hardware target is ≥ 50 Hz
Maximum following speed	0.8 m/s (unloaded)	Matches average comfortable shopping walking pace
Battery operational life	≥ 3 hours continuous	Must cover a full extended shopping session without recharge
System boot-up time	<15 seconds	Acceptable wait time at cart collection point
Weight measurement accuracy	± 200 g	Sufficient for triggering load-based speed adjustments reliably

B. Safety Requirements

Safety is the highest priority non-functional requirement. The cart operates in a shared human environment and must never cause harm. The following safety requirements are mandatory for deployment:

- The cart shall never exceed 1.2 m/s under any operating condition.
- The emergency stop mechanism shall operate via a hardware relay, independent of software — cutting motor power even if the Raspberry Pi crashes.
- The cart shall detect and stop for obstacles lower than 20 cm from the floor (child feet, bags) using IR sensors.
- The system shall prevent motion when basket load exceeds 40 kg.
- If the UWB link to the user remote is lost for more than 5 seconds, the cart shall decelerate to a stop and revert to IDLE mode.
- The cart shall not operate on slopes greater than 10° without triggering a low-speed safe mode.
- All exposed electrical components shall be rated IP42 minimum for indoor retail environments.

C. Reliability and Availability

The Smart Cart is a consumer-facing product deployed in a commercial setting. Reliability failures have direct business and safety consequences.

- Mean Time Between Failures (MTBF): The system shall operate for a minimum of 500 continuous shopping sessions before requiring scheduled maintenance.
- The software watchdog timer shall recover from non-critical ROS 2 node crashes within 2 seconds without user intervention.
- Battery management shall provide at least 60 seconds of safe operation warning before critical shutdown.
- All sensor failures shall be detected within one processing cycle (25 ms) and trigger a safe-mode response.

D. Usability Requirements

The system must be accessible to all user classes without training, including elderly shoppers and those unfamiliar with technology.

- The physical remote provides three buttons: **Follow** (activate follow-me), **Stop** (emergency stop), and **Idle** (pause and hold position) — no touchscreen, no app, no setup.
- LED status colors shall follow universally recognized conventions: green = safe/active, yellow = caution, red = stop/fault.
- The remote shall weigh no more than 80 g and fit comfortably in one hand.
- A new user shall be able to begin using the cart within 30 seconds of receiving the remote, with no instruction beyond the LED color guide on the remote lanyard.

E. Maintainability and Serviceability

Mall operations staff must be able to maintain a fleet of Smart Carts with minimal specialist expertise.

- The battery pack shall be hot-swappable — replaceable in under 3 minutes without tools.
- All sensor modules shall be individually replaceable without disassembling the main control enclosure.
- The system shall log all fault events with timestamps to an onboard SD card for post-session review by technical staff.
- Firmware updates shall be deployable over USB without requiring disassembly of the cart.

F. Security Requirements

Although Smart Cart does not handle personal data or payment information, the remote-to-cart communication channel must be secured to prevent malicious interference in a busy retail environment.

- The UWB pairing between remote and cart shall use a unique session key generated at each power-on, preventing cross-cart interference or hijacking.
- No external network connectivity (WiFi, Bluetooth) shall be active during normal operation — the system is fully self-contained.

- Firmware shall only accept signed update packages to prevent unauthorized code execution.

V. EXPERIMENTAL RESULTS

A. Simulation Setup

All experiments were conducted in Gazebo Harmonic with ROS 2 Jazzy on a desktop PC running Ubuntu 22.04 (Intel Core i7, 16 GB RAM). The simulated environment replicates an indoor retail aisle: two static shelf models at ± 1.5 m from the centre line, a static orange obstacle box ($0.3 \times 0.3 \times 0.5$ m) at position (2.0, 0.6) m, and a diff-drive person model that is driven via keyboard (W/A/S/D) at approximately 0.4 m/s per trial. Keys 1/2/3 on the same keyboard simulate the handheld remote sending STOP/FOLLOW/IDLE commands to the navigation state machine.

Each test was repeated five times. Metric data were captured by echoing `/uwb/distance`, `/cmd_vel`, and `/nav/status` during each run. All timing values were derived from ROS 2 message timestamps.

B. UWB Localisation Accuracy

The `uwb_simulator_node` generates four-anchor noisy range measurements ($\sigma = 0.08$ m, bias = 0.02 m) from Gazebo ground-truth odometry. The `follow_me_node` runs least-squares trilateration followed by a Kalman2D filter ($Q = 0.05\mathbf{I}$, $R = 0.3\mathbf{I}$, 10-cycle warm-up at 20 Hz = 0.5 s). To evaluate localisation accuracy, the person model was held stationary at five known distances and 100 UWB estimates (5 s) were collected per position.

Table XI
UWB LOCALISATION ACCURACY — RAW VS. KALMAN-SMOOTHED
(GAZEBO SIMULATION, AUTHOR DEVELOPED)

True (m)	Dist	Trials	Raw (m)	Error	Kalman Error (m)
1.0		5	0.11 $\pm$ 0.05		0.07 $\pm$ 0.03
1.5		5	0.13 $\pm$ 0.06		0.08 $\pm$ 0.03
2.0		5	0.14 $\pm$ 0.07		0.09 $\pm$ 0.04
3.0		5	0.17 $\pm$ 0.08		0.11 $\pm$ 0.04
4.0		5	0.21 $\pm$ 0.09		0.13 $\pm$ 0.05

The Kalman filter reduced position estimation error by approximately 35–40 % compared to raw trilateration across all tested distances. All smoothed errors remain well within the <0.30 m UWB positioning target specified in the non-functional requirements (Section IV). Estimation error increases with distance, which is consistent with the worsening trilateration geometry as the person moves further from the anchor array.

C. Follow-Me Tracking Performance

Three follow-me scenarios were evaluated with FOLLOW mode active (key 2 pressed). The person model was driven at approximately 0.4 m/s (comfortable walking pace). Following distance was sampled continuously from `/uwb/distance` at 20 Hz throughout each trial.

During straight-line following, the cart maintained a mean following distance of 1.04 ± 0.08 m — within the 1.0 ± 0.15 m

Table XII
FOLLOW-ME TRACKING RESULTS — GAZEBO SIMULATION (AUTHOR DEVELOPED)

Scenario	Trials	Mean (m)	Dist	Result
Straight-line (10 m)	5	1.04 $\pm$ 0.08		Pass
90° turn following	5	1.11 $\pm$ 0.13		Pass
Stop-and-resume	5	1.03 $\pm$ 0.07		Pass

acceptance window defined by `FOLLOW_DIST_M = 1.0` and `DEAD_ZONE_M = 0.15`. During 90° turns the cart re-acquired the target bearing within 1.5 s of the person changing direction, with overshoot bounded at ≈ 0.13 m (within the dead zone). In stop-and-resume trials, the cart held its position while the person was stationary and resumed following within one control cycle (≤ 50 ms) when motion restarted.

D. Obstacle Avoidance Performance

The `obstacle_stop_node` monitors the forward $\pm 45^\circ$ LiDAR arc and classifies returns into three proximity zones. Tests used the static orange obstacle box placed at varying distances along the cart's path. The person model was not in the path during these trials to isolate obstacle behaviour.

Table XIII
OBSTACLE AVOIDANCE ZONE VERIFICATION — GAZEBO SIMULATION
(AUTHOR DEVELOPED)

Test Scenario	Trials	Success	Stop Dist (m)
CAUTION zone (<1.5 m, 80% speed)	5	5/5	1.42 $\pm$ 0.06
SLOW zone (<0.8 m, 50% speed)	5	5/5	0.74 $\pm$ 0.05
Emergency stop (<0.3 m, 0% speed)	5	5/5	0.27 $\pm$ 0.03
Follow-Me with static box in path	5	5/5	No collision

All five emergency stop trials triggered a full velocity zero at a mean obstacle distance of 0.27 m, correctly within the `EMERGENCY_STOP_M = 0.30` m threshold. Speed reduction to 50 % and 80 % was verified by comparing `/cmd_vel` linear velocity to `/cmd_vel_raw` during SLOW and CAUTION zone approaches. No false emergency stops were triggered by the followed person model in FOLLOW mode, confirming that the `FOLLOW_MODE_MIN_OBSTACLE_M = 0.40` m filter correctly excludes the person from obstacle classification.

E. System Latency and Responsiveness

End-to-end control pipeline latency was measured using ROS 2 message timestamps across the two-hop path: `/uwb/distances` $\rightarrow$ `follow_me_node` $\rightarrow$ `obstacle_stop_node` $\rightarrow$ `/cmd_vel`.

The end-to-end latency of ≈ 5 ms is well within the 30 ms sensor-to-decision requirement specified in Section IV. The

Table XIV
CONTROL PIPELINE LATENCY — GAZEBO SIMULATION (AUTHOR DEVELOPED)

Pipeline Stage	Mean (ms)	Max (ms)
UWB pub → follow_me output	3.1	7.4
follow_me → obstacle output	1.8	4.2
End-to-end (UWB → /cmd_vel)	4.9	11.6
Control loop period (20 Hz nom.)	50.0	52.3

20 Hz control loop ran with <2.3 ms jitter across all measured cycles, confirming stable timer scheduling under Gazebo simulation load.

F. Discussion

The simulation results confirm that all three core behaviours — UWB localisation, follow-me tracking, and obstacle avoidance — operate within their design specifications.

The proportional follow-me controller ($K_P = 0.6$, $v_{max} = 0.8$ m/s) with a 0.15 m dead zone provides stable tracking during straight-line following. The slightly higher overshoot observed during 90° turns (mean distance 1.11 m vs. target 1.0 m) is attributed to the angular controller lag and the Kalman filter’s 10-cycle warm-up period when the person’s bearing changes rapidly. This remains within the 0.15 m acceptance window and does not cause unsafe cart behaviour.

The two-stage velocity pipeline (/cmd_vel_raw → obstacle_stop_node → /cmd_vel) successfully decouples follow-me logic from obstacle safety, allowing both to be tuned independently. The FOLLOW mode obstacle filter (ignoring returns <0.40 m) proved essential: without it, the person’s own LiDAR return would trigger continuous false emergency stops during normal following.

Observed limitations include: (1) mild oscillation at the start of each FOLLOW session during Kalman warm-up (10 cycles, ≈ 0.5 s); (2) UWB trilateration geometry degrades slightly at distances >3 m when the person moves close to the anchor boresight, causing a transient increase in position error; and (3) the teleop person model moves deterministically, which does not fully replicate unpredictable human gait. These limitations are addressed in the future improvements discussed in Section VIII.

G. Technical Feasibility

All core technologies employed in the Smart Cart system are commercially available, well-documented, and have been demonstrated in comparable robotics research applications. The Raspberry Pi 4B running ROS 2 is a proven platform for mobile robotics. UWB-based indoor positioning achieves sub-30 cm accuracy in retail-like environments, as documented in recent literature. Omni-wheel holonomic platforms are a mature technology used extensively in service robots. The primary technical risks are UWB multipath interference in

highly metallic shelf environments and real-time control latency over the Arduino-to-RPi serial bridge — both of which have well-known mitigation strategies (antenna diversity and DMA serial respectively).

H. Economic Feasibility

The following table provides a component-level bill of materials (BOM) with estimated unit costs for a single prototype unit. Costs are based on current retail pricing from distributor catalogues (RS Components, Mouser, Amazon UK) as of 2025.

Table XV
BILL OF MATERIALS — SINGLE PROTOTYPE UNIT (AUTHOR DEVELOPED)

Component	Qty	Unit (£)	Total (£)	Source
Raspberry Pi 4B (4GB)	1	55	55	RS Components
Arduino Mega 2560	1	38	38	Arduino Official
DWM1000 UWB Module	5	18	90	Mouser
DC Motor + Omni Wheel Set	4	22	88	RobotShop
L298N Dual H-Bridge Driver	2	8	16	Amazon UK
HC-SR04 Ultrasonic Sensor	2	4	8	Amazon UK
Sharp IR Distance Sensor	4	9	36	RS Components
50kg Bar Load Cell + HX711	4	7	28	Amazon UK
24V 20Ah LiPo Battery + BMS	1	95	95	Hobbyking
Buck Regulators (XL4016)	2	6	12	Amazon UK
RGB LED Strip (WS2812B)	1	12	12	Amazon UK
Remote PCB + Buttons + Case	1	25	25	Custom
Frame Mounts + Wiring + Misc	1	40	40	Various
TOTAL PROTOTYPE COST			£543	

For context, a standard commercial shopping cart costs approximately £80–£150. The Smart Cart prototype at £543 represents a $\sim 4\times$ cost multiplier for a single unit. At scale (500+ units), electronics costs reduce significantly through bulk purchasing and custom PCB fabrication, with an estimated production cost of £280–£320 per unit. Given the operational benefits — reduced staff assistance needs, reduced cart damage, and improved shopper retention — a payback period of 18–24 months is estimated for large-format retail deployments.

I. Environmental Feasibility

The system is designed for operation exclusively within controlled indoor retail environments. Key environmental constraints and their mitigations are summarized below:

VI. TESTING SPECIFICATIONS

Testing is structured in four progressive phases, from unit-level hardware verification through to full-scenario integration

Table XVI
ENVIRONMENTAL FEASIBILITY ASSESSMENT (AUTHOR DEVELOPED)

Environmental Factor	Constraint	Mitigation
Floor Surface	Smooth, level (max 3° ramp)	Omni-wheels handle minor slopes; ramp detection via IMU
Lighting	Moderate indoor (>200 lux)	UWB and ultrasonic are light-independent
Crowd Density	Up to 2 people/m ²	DWA planner with conservative speed limits
RF Interference	Dense WiFi / BLE environment	UWB at 3.5–6.5 GHz; minimal overlap with WiFi
Temperature	15–30°C standard retail	All components rated for 0–70°C operation

tests in a simulated retail environment. Each test specifies acceptance criteria that must be passed before progressing to the next phase.

A. Phase 1 — Component Unit Tests

Table XVII
PHASE 1 — COMPONENT UNIT TESTS (AUTHOR DEVELOPED)

ID	Description	Method	Acceptance Criteria
T1.1	UWB anchor ranging accuracy	Static tag at known positions (1 m, 2 m, 5 m)	Error < ± 30 cm at all ranges
T1.2	Ultrasonic distance accuracy	Target board at 0.3, 0.5, 1.0, 2.0 m	Error < ± 3 cm; latency < 25 ms
T1.3	IR sensor low-obstacle detection	Obstacle at 10, 40, 80 cm floor level	Detected at all ranges within spec
T1.4	Weight sensor calibration	Known masses: 0, 5, 10, 20, 40 kg	Error < ± 200 g across range
T1.5	Motor direction and speed	PWM sweep 0–100% all 4 motors	No stall; encoder counts consistent
T1.6	Emergency stop response time	Remote X button press to motor stop	Full stop within 200 ms
T1.7	LiDAR 360° FoV Check	Place cart in clear area and observe ROS 2 /scan topic while rotating; verify basket frame occlusion	No blind spots or data gaps; static chassis frame must be filtered out
T1.8	Camera Visual Odometry (VO) drift	Move cart using only camera with encoder disabled	Camera-based movement must have minimal drift (< 10–15 cm) over measured distance

B. Phase 2 — Subsystem Integration Tests

C. Phase 3 — Autonomous Navigation Tests

D. Phase 4 — Full Scenario / Performance Tests

VII. RISK ASSESSMENT

The following risk register identifies key technical, operational, and safety risks associated with the Smart Cart system,

Table XVIII
PHASE 2 — SUBSYSTEM INTEGRATION TESTS (AUTHOR DEVELOPED)

ID	Description	Method	Acceptance Criteria
T2.1	ROS 2 topic publish rates	ros2 topic hz monitoring all nodes	UWB topics at ≥ 20 Hz; control loop at ≥ 20 Hz; nav status at ≥ 2 Hz
T2.2	Serial bridge latency	Timestamp sensor data vs. ROS topic	End-to-end latency < 30 ms
T2.3	UWB + Kalman filter stability	Move person around cart perimeter 3× using teleop_person_wand	No position jumps > 20 cm; Kalman-smoothed estimate converges after warmup
T2.4	Obstacle zone classification	Approach cart with obstacle at varying speeds	Correct zone output at all thresholds
T2.5	Speed adaptation to load	Test at 0, 10, 20, 40 kg basket load	Speed reduces proportionally rules
T2.6	Watchdog timer fault	Kill RPi process; observe Arduino	Motor cut within 500 ms; red LED flash

Table XIX
PHASE 3 — AUTONOMOUS NAVIGATION TESTS (AUTHOR DEVELOPED)

ID	Description	Method	Acceptance Criteria
T3.1	Follow-me straight line	User walks 10 m corridor; cart follows	Cart maintains 1.0 m $\pm$ 0.15 m following distance
T3.2	Follow-me with turns	User makes 90° left and right turns	Cart completes turn within 1.5 s with no collision
T3.3	Obstacle avoidance — static	Static box placed in cart path mid-follow	Cart re-routes without stopping > 2 s
T3.4	Obstacle avoidance — dynamic	Person walks across cart path at 1.0 m/s	No contact; recovery to follow-path within 3 s
T3.5	Narrow aisle navigation	Aisle width of 0.9 m (cart is 0.7 m wide)	Cart passes without wall contact

Table XX
PHASE 4 — FULL SCENARIO PERFORMANCE TESTS (AUTHOR DEVELOPED)

ID	Description	Method	Acceptance Criteria
T4.1	Full 2-hour battery endurance	Continuous operation, avg 0.5 m/s, 15 kg load	Battery > 20% at 2 hours; no fault interrupts
T4.2	Simulated shopping route	User completes 20-min mall route with stops	$\geq 95\%$ successful follow-me engagement time
T4.3	Crowded environment (10 people)	10 actors walking in simulated mall space	Zero collisions; ≤ 3 emergency stops triggered
T4.4	User stop-and-shop behaviour	User pauses for 30 s, resumes walking	Cart holds position; resumes follow-me within 1 s
T4.5	Repeated E-Stop recovery	10× stop/resume cycles in 5 minutes	All recoveries successful; no system fault logged

along with their likelihood, severity, and proposed mitigation strategies.

Table XXI
RISK REGISTER — SMART CART SYSTEM (AUTHOR DEVELOPED)

Risk	Like.	Sev.	Level	Mitigation
UWB multipath errors in metallic shelving	Med	High	HIGH	Deploy 4-anchor array; use EKF smoothing; test in representative environment
Cart collides with shopper leg or heel	Low	High	HIGH	Multi-zone obstacle detection; hardware E-stop relay; speed limits enforced in software
Battery depletion mid-session	Low	Med	MEDIUM	BMS low-battery alert at 20%; geofenced return-to-base at 15%
RPI thermal throttle under load	Med	Med	MEDIUM	Heatsink + 5 V fan on RPi; monitor CPU temp via ROS 2 diagnostics node
Serial communication dropout between RPi and Arduino	Low	High	HIGH	Watchdog timer on Arduino; automatic re-connect in sensor_bridge node
User walks faster than cart max speed	Med	Low	LOW	Cart switches to stationary follow mode; user notified via LED; max speed can be tuned
Children or pets triggering false obstacle stops	High	Low	MEDIUM	IR sensor height calibrated to ignore floor-level noise; adjustable stop thresholds
Cart remote lost or stolen	Med	Med	MEDIUM	Bluetooth pairing lock; remote-loss timeout (60 s idle → auto park mode)
Omni-wheel performance on wet floors	Low	Med	MEDIUM	Speed reduction on incline (IMU); non-slip wheel treads; wet-floor warning in store policy
ROS 2 node crash causing uncontrolled motion	Low	High	HIGH	Watchdog timer; hardware relay E-stop; all motor commands time out if no update >300 ms

Risk levels are assessed using a standard 3×3 likelihood-severity matrix. HIGH risks require mandatory mitigation prior to deployment testing. MEDIUM risks require mitigation prior to full public operation. LOW risks are monitored and reviewed at each testing phase.

VIII. FUTURE IMPROVEMENTS

The current Smart Cart specification represents a first-generation system designed for practical deployment with commercially available components. As the system matures through real-world testing and user feedback, several enhancements are anticipated to improve capability, user experience, and commercial viability.

Of these enhancements, the Computer Vision Upgrade and Automatic Return-to-Base represent the highest-priority upgrades for a second-generation prototype. The Computer

Table XXII
PROPOSED FUTURE ENHANCEMENTS (AUTHOR DEVELOPED)

Enhancement	Description	Expected Benefit
Shopping App Integration	Bluetooth link to shopper's phone to display shopping list and aisle navigation	Transforms cart into a personalised shopping assistant
Automatic Return-to-Base	When session ends and remote is docked, cart navigates autonomously back to charging station	Reduces staff workload; keeps carts charged and organised
Multi-Cart Fleet Management	Central dashboard for mall operators to monitor battery, location, and fault status of all carts	Improves operational efficiency at scale
Computer Vision Upgrade	Replace UWB tracking with camera-based human pose estimation (e.g. MediaPipe)	Removes need for user to carry remote; fully hands-free operation
Voice Feedback	Add a small speaker for audible alerts in noisy environments where LEDs may be missed	Improves accessibility for visually impaired shoppers

Vision Upgrade directly addresses the primary user-experience limitation of the current design — the requirement for the user to carry a UWB remote tag. The Return-to-Base feature addresses the operational limitation of requiring staff to manually collect and dock carts between sessions.

IX. CONCLUSION

This paper presented the design, simulation, and evaluation of an Autonomous Smart Cart system built on ROS 2 Jazzy and Gazebo Harmonic. The system integrates a UWB-based localisation pipeline, a 2D Kalman filter, a proportional follow-me controller, and a LiDAR obstacle safety layer to enable a differential-drive cart to autonomously track a user within a simulated indoor retail environment.

Simulation results demonstrated that the implemented system meets all primary functional objectives. The Kalman-smoothed UWB localisation achieves a position error of <0.13 m across the 1–4 m operating range, well within the 0.30 m accuracy target. The follow-me controller maintains a following distance of 1.0 ± 0.15 m during both straight-line and turning scenarios. The obstacle safety layer correctly enforces three-zone proximity speed scaling and triggers emergency stops at the specified 0.30 m threshold, with no false stops caused by the followed person due to the FOLLOW mode obstacle filter. End-to-end control latency of ≈ 5 ms satisfies the 30 ms non-functional requirement.

The modular ROS 2 architecture — separating localisation, follow-me control, navigation state management, and obstacle safety into independent nodes connected via publish-subscribe topics — proved effective for iterative testing and independent tuning of each subsystem. The two-stage velocity pipeline in particular provides a clean separation between navigation intent and safety enforcement.

Key limitations of the current simulation include the use of a teleop-controlled person model (rather than autonomous human

motion), a purely proportional controller (which introduces overshoot during rapid directional changes), and the absence of SLAM-based global localisation. These form the basis for the future improvements described in Section VIII, with computer-vision-based human tracking and autonomous return-to-base identified as the highest-priority second-generation upgrades.

Overall, the Gazebo simulation validates the core system architecture and demonstrates that UWB-guided follow-me with layered LiDAR obstacle avoidance is a viable and safe approach for autonomous indoor cart systems. The codebase, launch configuration, and simulation world are publicly available on GitHub for reproduction and further development.

APPENDIX

This appendix has been updated to reflect AI tool usage during CW2 simulation implementation and report drafting.

In accordance with Middlesex University’s assessment guidelines and the module requirement to declare AI usage, this appendix discloses how and where generative AI tools were used in the preparation of this document.

A. How AI Was Used

The team used Claude (Anthropic) as a writing and structuring aid throughout the preparation of this report. AI assistance was used in the following ways:

Table XXIII
AI USAGE DISCLOSURE BY SECTION (AUTHOR DEVELOPED)

Section	AI Role	Human Contribution
Introduction and Problem Statement	Draft structure and academic language polishing	Original problem observations, research citations, and framing
Hardware Specifications (Section III)	Generating specification tables with realistic component values	Team verified all components against real datasheets and design diagram
Software Architecture (Section III)	Drafting ROS 2 node graph and communication protocol tables	Team defined the actual node structure, sensor choices, and control logic
Feasibility and Cost Analysis (Section V)	Generating BOM table with estimated costs	Team reviewed all prices against current supplier listings and adjusted
Testing Specifications (Section VI)	Drafting test case structure and acceptance criteria	Team defined test scenarios based on actual design goals
Risk Assessment (Section VII)	Populating risk register template	Team identified risks based on real design knowledge and class content

B. Human Oversight and Verification

All AI-generated content was reviewed, edited, and verified by team members before inclusion in this report. The core design decisions — including the choice of UWB positioning over camera-based tracking, the selection of omni-wheels, the Raspberry Pi + Arduino dual-processor architecture, and the remote-control interface — were developed independently by the team through group discussion, class knowledge, and the

design sketch produced during the ideation phase. AI was used as a productivity and language tool, not as a design decision-maker.

Team members take full academic responsibility for the content of this document.

C. AI Tools Used

- Claude (Anthropic) — document drafting, structuring, and table generation
- No AI-based image generation tools were used; all diagrams are author-developed

This section documents the design thinking process used by the team to arrive at the final Smart Cart concept. Design thinking was applied across five stages: Empathize, Define, Ideate, Prototype, and Test. Each stage shaped key decisions about the system’s hardware configuration, user interaction model, and safety approach.

D. Empathize — Understanding the Problem

The team began by observing shopping behavior in a large retail mall environment. Key pain points identified through observation and informal user interviews included:

- Elderly shoppers struggled to push heavily loaded carts, especially on slight inclines near entrances.
- Parents with young children frequently lose control of carts when distracted.
- Narrow aisles in peak hours caused collisions between carts and other shoppers, particularly at ankle and heel height.
- Cart return was often neglected, causing trolleys to roll freely in car parks.

These direct observations motivated the core design goal: a cart that removes physical burden from the user while actively preventing collisions in dynamic environments.

E. Define — Problem Statement

Based on the empathy stage, the team defined the central problem as:

“Shoppers in crowded retail environments need a smarter cart that follows them autonomously, detects obstacles, and adapts to load — so they can shop safely without physical strain.”

F. Ideate — Design Concepts Considered

The team generated three competing design concepts before converging on the final solution:

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Table XXIV
DESIGN CONCEPTS CONSIDERED DURING IDEATION (AUTHOR
DEVELOPED)

Concept	Description	Why Rejected / Adopted
Concept A	GPS-guided outdoor cart with motorized wheels	GPS unreliable indoors; rejected for indoor-only scope
Concept B	Camera-only follow system (no UWB)	Camera occlusion in crowds caused unreliable tracking; rejected
Concept C (Chosen)	UWB-based follow-me with omni-wheels, ultrasonic + IR sensors	Most reliable indoors; handles occlusion; low cost; ADOPTED

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